

1. Executive Summary & Direct Answer

Primary Finding:

Across 10,216 platform-attributed posts in the Data Vortex ecosystem, **Instagram** generates the highest average total engagement per post at **3,669.38 interactions** (comprising 2,500.93 likes, 1,040.84 shares, and 499.80 comments). Overall platform performance is tightly clustered, with only a 2.96% spread separating the top platform (Instagram) from the lowest (Twitter at 3,563.75).

2. Official Platform Interaction Benchmark Table

Platform	Post Count	Avg Likes	Avg Shares	Avg Comments	Avg Total Engagement
Instagram	1,989	2,500.93	1,040.84	499.80	3,669.38
Reddit	2,031	2,488.11	1,002.23	511.18	3,647.47
YouTube	2,073	2,517.77	1,011.83	504.38	3,638.03
Facebook	2,074	2,528.86	984.17	506.94	3,631.00
Twitter	2,049	2,437.69	1,005.39	506.13	3,563.75

3. Key Quantitative Findings & Channel Dynamics

- **Viral Sharing Leadership (Instagram):** Instagram's top rank is propelled primarily by viral amplification, leading all platforms in average shares at **1,040.84** (+5.76% higher than Facebook's 984.17).
- **Conversational Depth (Reddit):** Reddit records the highest average comments per post (**511.18**), aligning with forum dynamics where textual debate and discussion exceed passive consumption.
- **Passive Consumption / Like Saturation (Facebook & YouTube):** Facebook achieved the highest average likes per post (**2,528.86**), followed closely by YouTube (**2,517.77**), reflecting high single-click engagement.
- **Benchmark Uniformity:** Mean total interactions exhibit extremely tight clustering across channels (mean ~3,630, standard deviation across platforms < 40). This uniform baseline suggests broad structural consistency across social channels in this corpus.

4. Methodological Limitations & Non-Causal Framing

- **Observational Association:** Differences between platforms represent descriptive population averages within this specific dataset. They do not demonstrate that publishing identical content on Instagram will cause higher engagement.
- **Unattributed Cohort:** 1,784 posts (14.87% of the total dataset) have unrecorded platform identifiers. While excluded here per official instructions, their average engagement (3,609.53) mirrors the overall mean, indicating missingness is missing completely at random (MCAR).
- **Missing Likes Attribution:** Approximately 15% of posts across each platform had unrecorded likes. As demonstrated in the Logic Explanation, Instagram's leading rank remains unchanged across all NULL-handling protocols.